



NIT W

HITACHI

SMART ENERGY MONITORING & PREDICTION SYSTEM

ARYAN CHOUHAN

DRISHTI JAIN

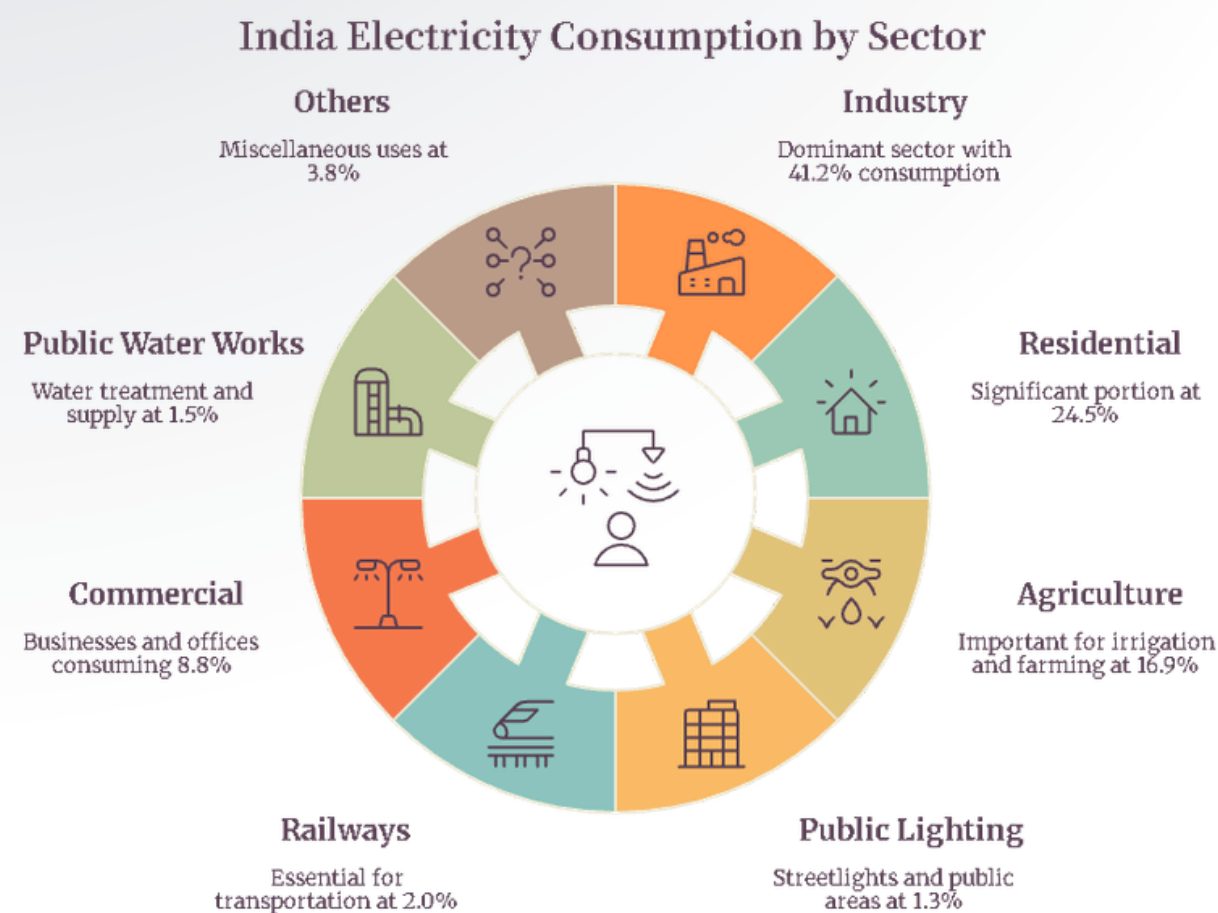
ADITYA KUMAR SAHOO

KRISHNA PANDEY

“Millions of households and industries consume electricity every day, yet they have no visibility into where the energy is actually being used.”

- **15.04% of electricity distributed in India was lost as AT&C losses in FY2024–25.**
- **1 in every 6 units of electricity is lost due to technical and commercial losses.**

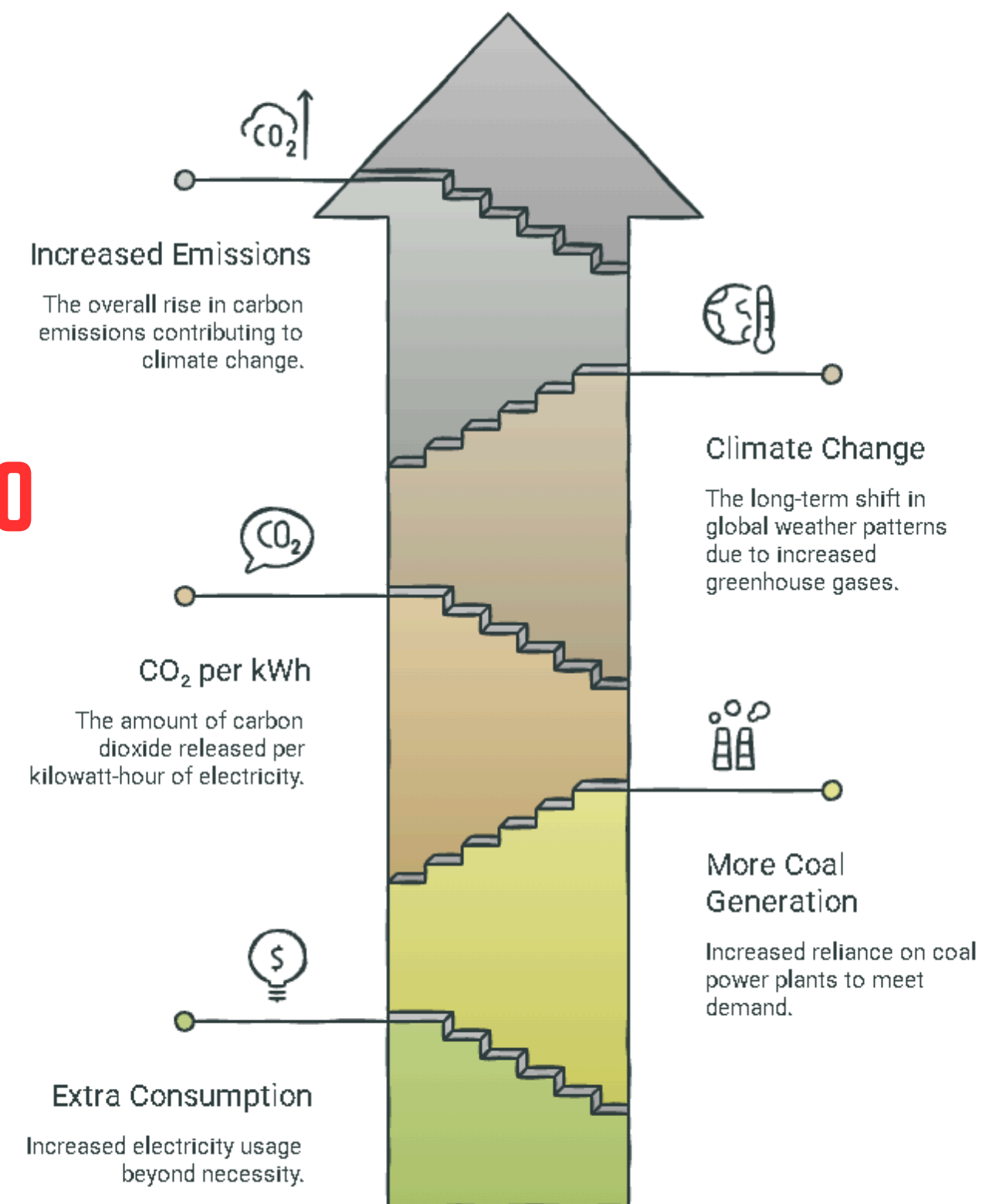
A team member's monthly electricity bill increased from ₹5,000 to ₹11,000 due to an undetected underground wiring fault that leaked current into the earth for several weeks.



EXPECTED BILL: ₹5,000
ACTUAL BILL: ₹11,000
HIDDEN LOSS: ₹6,000

Key Insight:
Hidden faults silently increase electricity bills.

Environmental Impact of Electricity Waste



EXISTING SOLUTION

- Smart meters measure total energy consumption only.
- Smart plugs require one device per appliance.
- Industrial EMS requires multiple CTs and sensors.
- Installation is complex and expensive.
- Most solutions depend on cloud platforms and subscriptions.
- Limited solutions provide real-time appliance-level monitoring.

Problems with Existing Solutions

- **Multiple sensors required**
- **One sensor per appliance/circuit**
- **High installation cost**
- **Difficult to retrofit existing buildings**
- **High maintenance**
- **Limited affordability for homes**

Characteristic	Technology	Sensors Required	Approx. Cost	Limitations
Schneider Electric EcoStruxure	IoT Energy Management	Multiple CTs & Smart Meters	₹50,000–₹5,00,000+	Expensive, industrial deployment
Siemens Sentron	Smart Energy Monitoring	Multiple Energy Meters	₹1–10 Lakhs	Complex installation
Energyly (India)	IoT Energy Monitoring	Multiple CT Sensors	₹50,000+	No appliance identification
Greeniee (India)	AI Energy Monitoring	Single Gateway	Enterprise Pricing	Commercial use only
Kimbal NILM (India)	AI + NILM	Single Clamp Device	Not Public	Proprietary solution
Enciser Emerald	Home Energy Monitor	Single CT Clamp	₹6,000	Shows total usage only
Sense (USA)	NILM + AI	Two CT Clamps	₹30,000–₹40,000	Not available in India
Emporia Energy	Smart Home Monitoring	8–16 CT Sensors	₹15,000–₹30,000	Multiple sensors required
Curb Energy	Smart Panel Monitoring	18–36 CT Sensors	₹70,000–₹1,20,000	High installation cost

● SMART ENERGY INFRASTRUCTURE



ऊर्जासेतु

URJASETU

Bridging India's energy future with intelligent, connected power.

Smart Grid



Clean Energy



AI Forecasting



Real-time Analytics

WHAT WE DID??



- SINGLE VOLTAGE & CURRENT SENSOR
- ESP32-BASED SMART MONITORING
- AI/ML-BASED APPLIANCE DETECTION (NILM)
- IOT CLOUD MONITORING
- LOW-COST PROTOTYPE (~₹2,500)

Parameter	Sensors Required	Installation	Cost	Appliance Detection	Scalability	Retrofit	Monitoring	Cloud Support	Maintenance
Existing Solutions	Multiple sensors / Smart plugs	Complex wiring	₹50,000– ₹2,00,000+	Multiple sensors per load	Limited	Difficult	Total energy only (many systems)	Optional / Paid	High
UrjaSetu	✓ Single Voltage + Single Current Sensor	✓ Simple plug-and-play	✓ ~₹2,500 (Prototype)	✓ AI/ML-based NILM	✓ Homes, Offices & Industries	✓ Works with existing wiring	✓ Appliance-level insights	✓ Real-time IoT Dashboard	✓ Low

HARDWARE SETUP

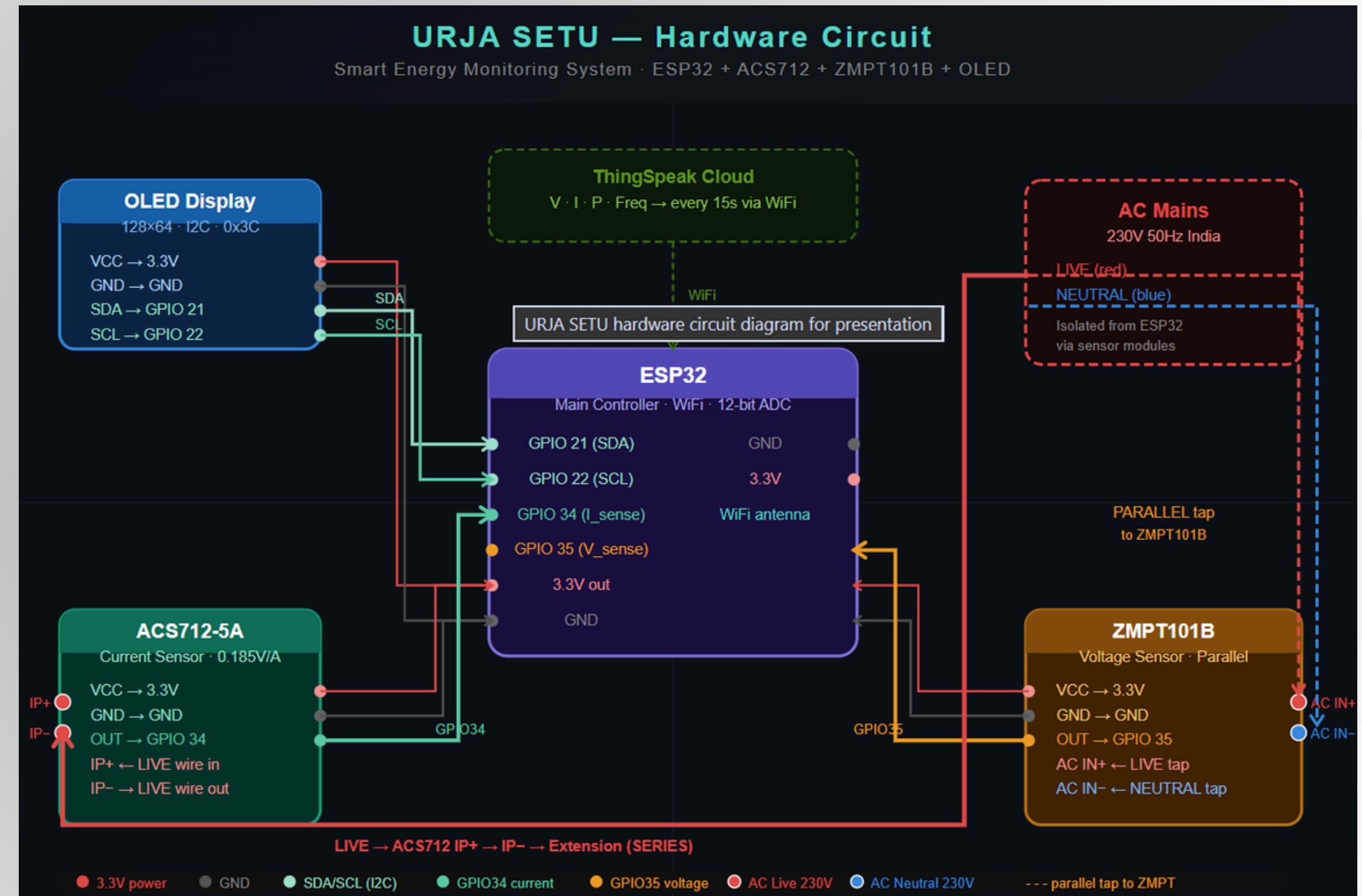


SYSTEM OVERVIEW

- EESP32 FOR PROCESSING & WI-FI COMMUNICATION
- REAL-TIME VOLTAGE & CURRENT SENSING
- LIVE DATA ON OLED & THINGSPEAK CLOUD

CLOUD INTEGRATION

- ESP32 UPLOADS ENERGY DATA TO THINGSPEAK VIA WI-FI
- REMOTE MONITORING & HISTORICAL DATA LOGGING



SAFETY NOTE

HIGH-VOLTAGE AC SIDE IS FULLY ISOLATED FROM THE ESP32'S LOW-VOLTAGE LOGIC THROUGH THE SENSOR MODULES

DATASET



⚠ BENCHMARK CONFLICT

PLAID vs OURS

A model trained on PLAID (US benchmark) would fail due to mismatch in physics and hardware.

Standard (PLAID)
120V / 60Hz

VS

Indian Mains
245V / 50Hz

Sensor Hardware

ESP32 + ACS712

Noise Profile

Real-world Mains

🗄 COLLECTION SPECS

Total Samples

1,824 Records

Scenarios

16 Combinations

Sampling Rate

4 Samples / Sec

Window Duration

90s Per Scenario

Multi-Label Binary: Resolves single, triplet, and "all-four-ON" states.

📡 8 FEATURE SIGNATURES

Combining Magnitude and Shape for unique load fingerprints.

RMS Voltage

RMS Current

Real Power

Apparent Power

Power Factor

Reactive Power

Crest Factor

Harmonic Dist.

⚡ SIGNATURE INSIGHTS

Appliance	I _{rms}	PF	CF
100W Bulb	0.43A	~1.0	~1.41
60W Bulb	0.26A	~1.0	~1.41
Charger	0.10A	0.55	>1.8

AIML METHODOLOGY



DATASET PROFILE

Total Samples	1,824 ($X \in \mathbb{R}^{1824 \times 8}$)
Scenario Count	16 Unique Scenarios
Sampling Rate	4 Samples/Second
Time Window	90s Per Scenario

Labels: 4 Binary Appliances (Label Matrix y)

PIPELINE DESIGN

- **Train/Test Split:** 80/20 Stratified
- **Preprocessing:** StandardScaler
- **Validation:** Bootstrap Sampling
- **Out-of-Bag (OOB):** ~37% Validation Set

Training Set Size: 1459 Samples

RF ARCHITECTURE

N-Estimators	200 Decision Trees
Split Features	$\sqrt{8} \approx 3$ Random Selection

$$\text{Gini}(S) = 1 - \sum_i p_i^2$$

MODEL BENCHMARKS

Random Forest	95.9%
kNN (k=5)	92.1%
MLP (Neural Net)	63.8%

DEPLOYMENT SPECIFICATIONS

The **Random Forest** was selected due to its superior exact-match accuracy and robustness against overfitting through random feature subsetting.

FINAL MODEL INSTANCE
MultiOutputClassifier
Hamming Loss: 0.0158

INDIAN SYSTEM



GOVERNANCE & POLICY

TOP TIER

Ministry of Power Electricity Act 2003
CERC / SERCs DSM Rules & Tariffs

Policy sets interstate/state bands and national power framework.

SYSTEM OPERATIONS

CONTROL

Grid-India NLDC + 5 RLDCs
Monitoring Real-time DSM Ops

Centralized visibility of national frequency and energy flow.

BALANCING TIER

ENFORCEMENT

Generators Over-injection Penalties
SLDCs / DISCOMs State Balancing & Penalties

Enforcing grid stability through Deviation Settlement (DSM).

END-USER VISIBILITY GAP

11kV Substation & Feeder

Visible to Grid-India, but lacks granular device-level insights.

SPARSE DATA

Household Consumer

Fixed slab tariffs with absolutely no real-time frequency visibility.

ZERO SIGNAL

THE DATA BLIND SPOT

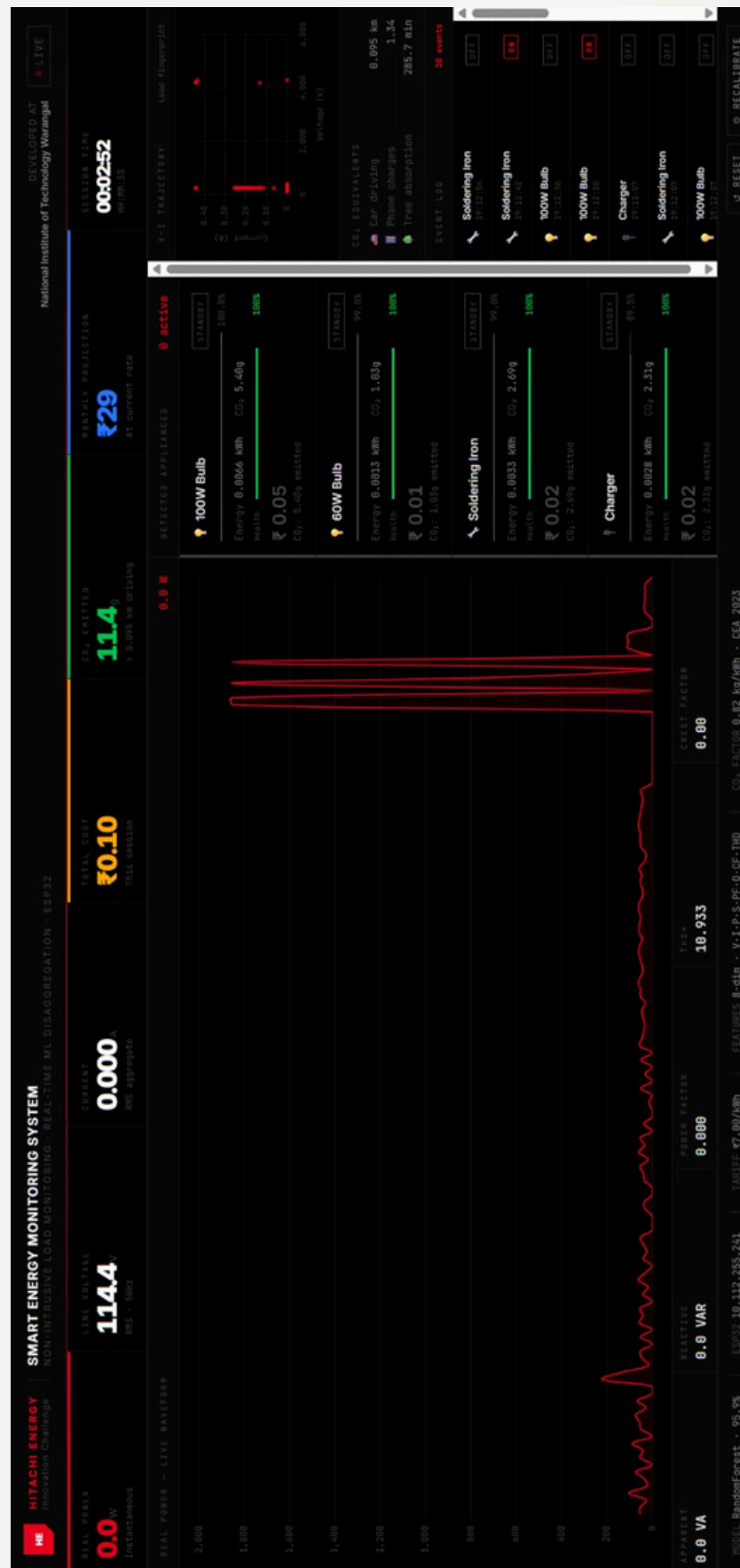
Below the substation level, the system operators lose all direct communication with individual consumer appliances.

System Optimization Component

OUR DEVICE

 Direct Consumer-to-Grid Intelligence

RESULT



Unified Electrical Monitoring

Random Forest Classifier

Identifies appliance combinations from sensor data.

Real-time Dashboard

Displays live electrical parameters in a centralized interface.

Monthly Bill prediction

Forecasts monthly electricity expenses for better planning.

Cost-Effectiveness

Reduces deployment costs with fewer sensors.

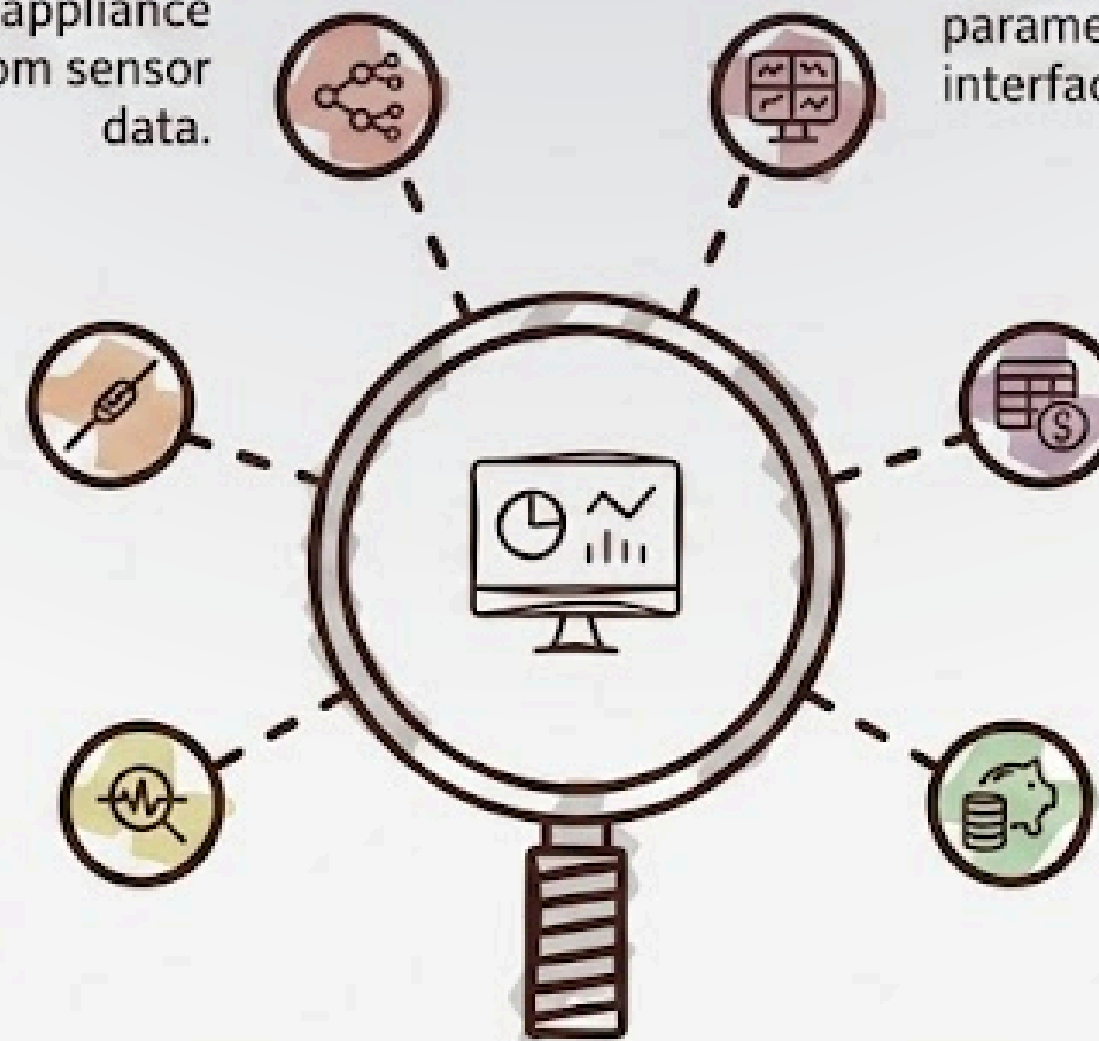
Efficient Energy Monitoring

Current Sensor

Detects electrical current flowing through appliances.

Voltage Sensor

Measures electrical voltage across multiple appliances.



FUTUTRE SCOPE



ADVANCED ANOMALY DETECTION
MONITORING LARGE HOUSEHOLD APPLIANCES FOR
EARLY FAILURE DETECTION



LARGER DATASET FOR AI
COLLECTING DIVERSE APPLIANCE DATA REAL HOUSEHOLDS



AUTOMATIC ALERTS
NOTIFYING USERS OF ABNORMAL ENERGY CONSUMPTION



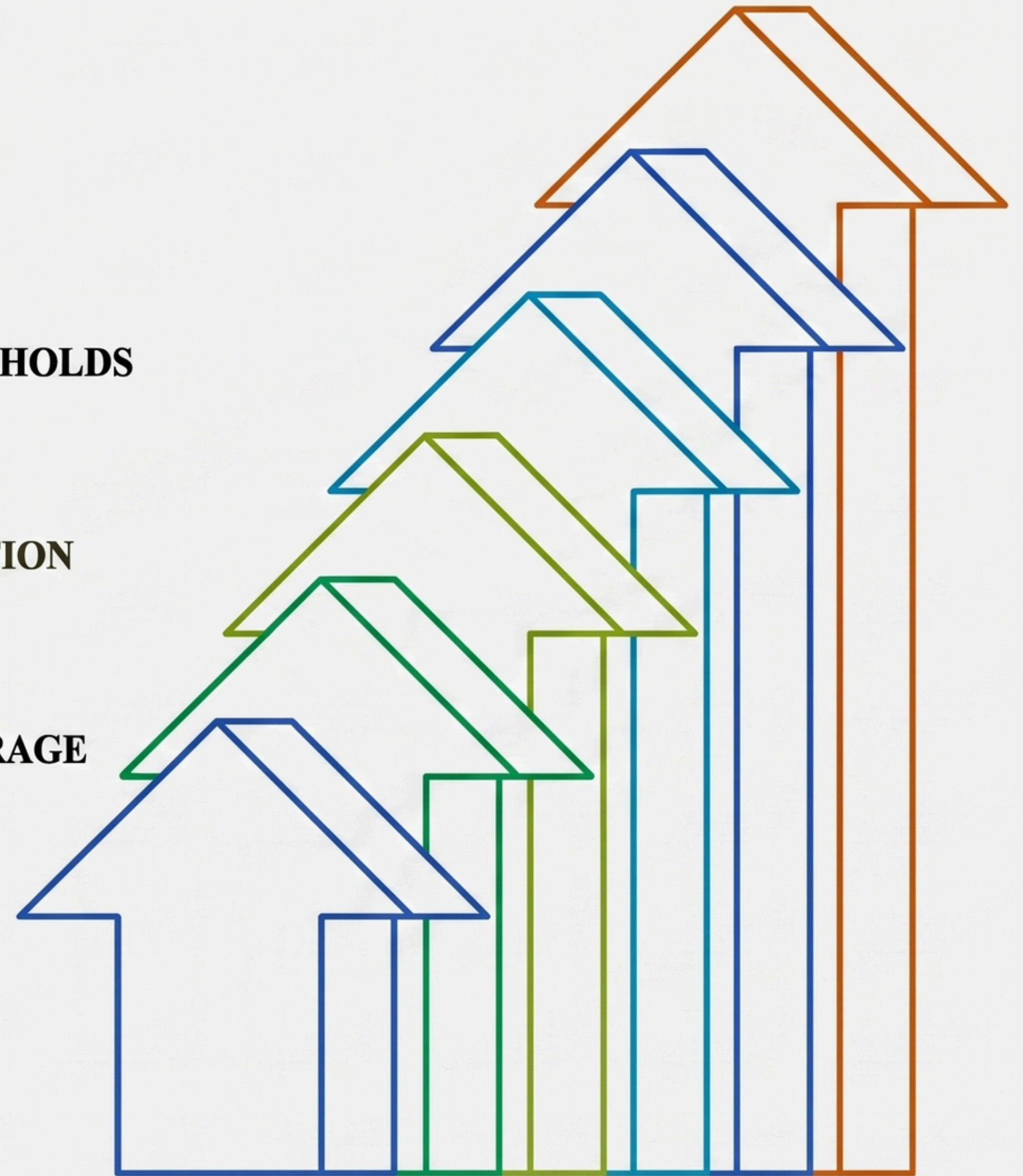
RENEWABLE ENERGY INTEGRATION
INTEGRATING WITH SOLAR PANELS AND BATTERY STORAGE



SMART HOME AUTOMATION
CONTROLLING APPLIANCES TO OPTIMIZE ENERGY
CONSUMPTION



CLOUD-BASED CONTINUOUS LEARNING
RETRAINING THE AI MODEL WITH NEW DATA



CONCLUSION



- AN AI-POWERED, IOT-ENABLED NILM SOLUTION FOR INTELLIGENT ENERGY MONITORING.
- AFFORDABLE, SCALABLE, AND EASY TO DEPLOY ACROSS MULTIPLE SECTORS.

“FROM A STUDENT INNOVATION TO A NATIONAL ENERGY SOLUTION—
URJASETU SEEKS THE MENTORSHIP, TECHNICAL EXPERTISE, AND INDUSTRIAL
COLLABORATION OF NIT WARANGAL AND HITACHI TO BUILD THE NEXT
GENERATION OF AI-DRIVEN SMART ENERGY SYSTEMS.”

Thank You

We Are Open for Questions !!!

